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PROGRAMMING WITH PYTHON

Lekha A

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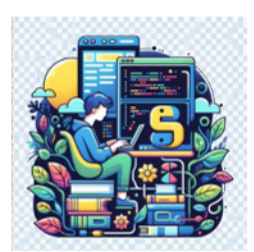
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Object-Oriented Programming (OOP) and Advanced Concepts

Constructors and Destructors

Lekha A

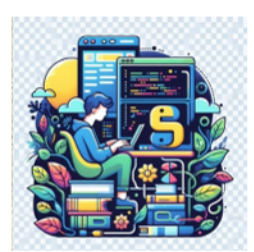
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Constructors

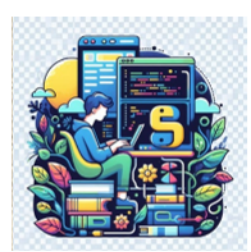
- In Python, a constructor is a special method that gets called when an object is created.
- It is used to **initialize** the object's attributes with default or given values.
- Constructor, as the name, suggests that is constructing something or, in other words, it is the first step of any task.
- Constructors are generally used for initializing an object.



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Constructors

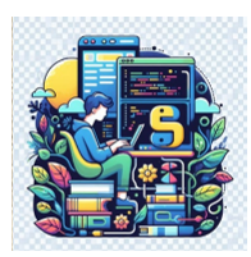
- In other languages, a method with the same name as the class is used to declare a constructor.
- But in Python, the `__init__()` method is called the constructor and is always called when an object of the class is created.
- The constructor is defined with the `__init__` method, which takes `self` as the first argument, followed by other arguments for initializing the object.



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Constructors

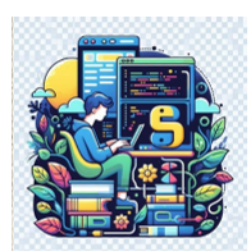
- There are two types of constructors in Python
 - Default Constructor
 - Parameterized Constructor



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Default Constructors

- The default constructor is a constructor that is automatically invoked even if the programmer doesn't make any constructors inside the class it creates.
- Its definition has only one argument which is the reference to the instance being constructed.



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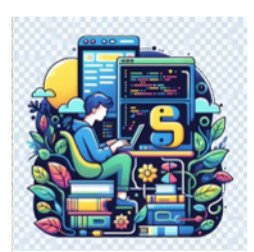
Default Constructors

```
class CrazyCoder:  
    def __init__(self):  
        self.code = "Pyton Programmer"  
    def print_code(self):  
        print(self.code)
```

```
obj = CrazyCoder()
```

```
obj.print_code()
```

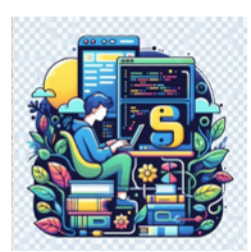
Pyton Programmer



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Parameterized Constructors

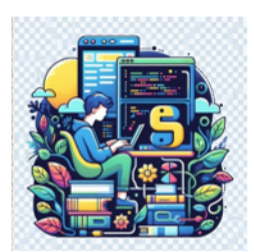
- A constructor with parameters passed as arguments is called a parameterized constructor.
- The parameterized constructor takes its first argument as a reference to the instance being constructed known as self and the rest of the arguments are provided by the programmer.



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Parameterized Constructors

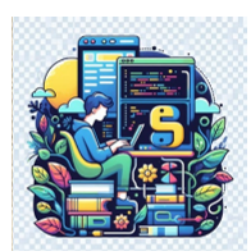
```
class Addition:
    num1 = 0
    num2 = 0
    answer = 0
    def __init__(self, f, s):
        self.num1 = f
        self.num2 = s
    def display(self):
        print("Num1 number = " + str(self.num1))
        print("Num2 number = " + str(self.num2))
        print("Addition of two numbers = " + str(self.answer))
    def calculate(self):
        self.answer = self.num1 + self.num2
```



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Destructors

- Destructors are the last code that runs when the object of a class gets destroyed.
- In python, destructors are not much needed as in C++ and Java.
 - Python by-default provides the garbage collector which handles the memory management.



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Destructors

```
class Animals:  
    def __init__(self):  
        print('The class called Animals is CREATED.')
```

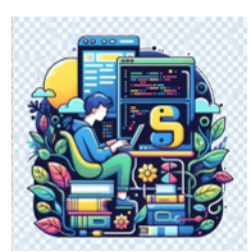
```
    def __del__(self):  
        print('The destructor is called for deleting the Animals.')
```

```
object = Animals()
```

The class called Animals is CREATED.

```
del object
```

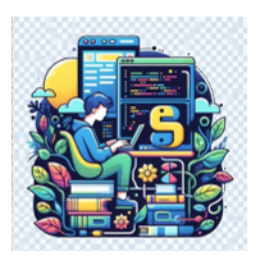
The destructor is called for deleting the Animals.



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Object Management

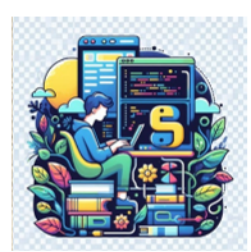
- Constructors and destructors are essential for proper object management in Python.
- They ensure that objects are initialized correctly and that any resources associated with them are released when they are no longer needed.
- This helps to prevent memory leaks and other resource-related issues.



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Garbage Collection

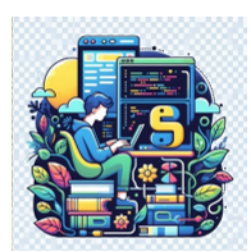
- Python employs a garbage collection mechanism to automatically reclaim memory occupied by objects that are no longer in use.
- This means that there is no need to explicitly call the destructor for every object.
- However, the `__del__()` method can be useful for performing any specific cleanup tasks that are not handled by the garbage collector.



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Real World Applications

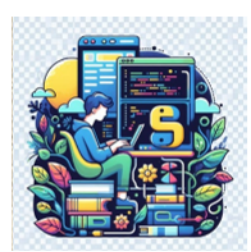
- Constructors and destructors are widely used in various Python applications, such as:
 - Managing database connections
 - Initializing a connection in the constructor and closing it in the destructor.



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Real World Applications

- Handling file operations
 - Opening a file in the constructor and closing it in the destructor.
- Releasing external resources
 - Acquiring resources in the constructor and releasing them in the destructor.



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Recap

- Constructors
 - Default Constructors
 - Parameterized Constructor
- Destructors
- Object Management
- Garbage Collection
- Real-World Applications



THANK YOU

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